

BookNest - Full Stack MERN Application

1. Introduction

BookNest is a full-stack e-commerce web application developed using the MERN stack (MongoDB, Express.js, React.js, Node.js). The platform allows users to browse books, register/login securely, manage cart items, place orders, and track order history.

2. Objectives

- To develop a complete MERN stack web application.
- To implement secure authentication and authorization using JWT.
- To design a responsive and user-friendly UI.
- To deploy a production-ready full-stack application.

3. System Architecture

The system follows a client-server architecture: Frontend (React) → Backend (Express API) → MongoDB Database. The frontend communicates with the backend using REST APIs. The backend handles authentication, business logic, and database interactions.

4. Technology Stack

Layer	Technology Used
Frontend	React.js, Vite, React Router, Axios
Backend	Node.js, Express.js
Database	MongoDB Atlas
Authentication	JWT, bcryptjs
Deployment	Render / Vercel

5. Features

- User Registration and Login with secure password hashing.
- JWT-based authentication and protected routes.
- Browse books with detailed view.
- Add to cart and manage cart items.
- Place orders and view order history.
- Role-based authorization (User/Admin).
- Responsive and modern UI design.

6. Authentication Flow

1. User registers → Password is hashed using bcrypt. 2. JWT token is generated upon successful login. 3. Token is stored in localStorage. 4. Protected routes validate the token via middleware. 5. Role-based authorization restricts admin-only routes.

7. API Endpoints

Method	Endpoint	Description
POST	/api/auth/register	Register new user
POST	/api/auth/login	Login user
GET	/api/auth/profile	Get logged-in user profile
GET	/api/books	Fetch all books
POST	/api/orders	Create new order
GET	/api/orders	Fetch user orders

8. Deployment

The backend is deployed on Render as a Web Service. The frontend is deployed on Render/Vercel as a Static Site. MongoDB Atlas is used for cloud database hosting.

9. Future Enhancements

- Payment gateway integration (Stripe/Razorpay).
- Email verification and password reset system.
- Search and pagination features.
- Admin dashboard for book management.

- Performance optimization and caching.

10. Conclusion

BookNest demonstrates a complete full-stack development workflow using the MERN stack. The project showcases authentication, REST API integration, responsive UI design, and deployment practices. It reflects strong understanding of modern web development principles.